



Time Series Analysis

Seminar 15: Review and exam preparation



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Acronyms used in this material (1/2)

- ▣ ACF: Autocorrelation Function
- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARCH-LM: ARCH Lagrange Multiplier (test)
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ DM: Diebold–Mariano (test)
- ▣ ECM: Error Correction Model
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin (test)
- ▣ LB: Ljung–Box (test)
- ▣ MA: Moving Average
- ▣ MAE: Mean Absolute Error
- ▣ MASE: Mean Absolute Scaled Error
- ▣ OLS: Ordinary Least Squares
- ▣ PACF: Partial AutoCorrelation Function

Acronyms used in this material (2/2)

- RMSE: Root Mean Squared Error
- ROBOR: Romanian Interbank Offer Rate
- RSS: Residual Sum of Squares
- SAR: Seasonal AutoRegressive term (EViews)
- SARIMA: Seasonal ARIMA (model)
- SE: Standard Error
- SES: Simple Exponential Smoothing
- SIGMASQ: SIGMA SQuared, the innovation variance (EViews)
- VAR: Vector AutoRegression
- VAT: Value Added Tax

Today's question and route

- **Question:** can you solve, on paper and on data, an exam problem from any core chapter?
 - ▶ this seminar comes **before** Lecture 15: the section “What you need today” is a formula sheet for [Chapters 0–10](#)
- **Route**
 - ▶ Part A: five exam-type problems on paper, with software output: ARIMA, identification, SARIMA, VAR and Granger, GARCH
 - ▶ Part B: three data tasks (industrial production, the BET, two 10-year yields), each with an interpretation question
 - ▶ Part C: a project idea and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

Exercise map

Task	Topic	Type	Chapters
A1	an ARIMA(0,1,1) by hand: moments, ACF, forecasts, exponential smoothing	Solved	0, 1, 3
A2	identifying the Romanian unemployment rate: ADF, KPSS, correlogram, Ljung–Box	Proposed	1–3
A3	SARIMA output for Romanian GDP and a forecast by hand	Proposed	4
A4	VAR output, Granger F from two RSS, a long-run effect	Proposed	6
A5	GARCH(1,1)- t output for the DAX and VaR 1%	Proposed	5
B1	Box–Jenkins for Romanian industrial production	Solved	3, 4
B2	volatility of the BET since 2015	Proposed	5
B3	the Romanian and German 10-year yields: cointegration	Proposed	3, 7
C1, C2	a forecast competition for inflation; audit an AI answer	Proposed	0–10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
unemployment rate, Romania (SA)	Eurostat (une_rt_m)	monthly	2005–2026
real GDP, Romania (not adjusted)	Eurostat (namq_10_gdp)	quarterly	2000–2026
HICP inflation and ROBOR 3M, Romania	Eurostat	monthly	2005–2026
industrial production, Romania (calendar adjusted)	Eurostat (sts_inpr_m)	monthly	2005–2026
DAX, BET	EODHD	daily	2000–2026
10-year yields, Romania and Germany	EODHD	monthly means of daily data	2008–2026

- Part A needs only a calculator: the output tables are on the slides; Part B uses the notebook
- Growth rates in %: $100\Delta \ln y_t$; annual rates: $100\Delta_{12} \ln y_t$ (monthly), $100\Delta_4 \ln y_t$ (quarterly); returns: $100\Delta \ln P_t$

What you need today (1/4): description, stationarity, ARMA

Quantity	Formula
SES, MASE	$\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$; MASE = MAE / MAE (seasonal naive, in sample)
ACF, band	$\hat{\rho}(h) = \hat{\gamma}(h)/\hat{\gamma}(0)$, $\pm 1.96/\sqrt{T}$
Ljung-Box	$Q^*(m) = T(T+2) \sum_{h=1}^m \hat{\rho}(h)^2 / (T-h) \sim \chi^2(m-k)$, k = ARMA parameters
AR(1), MA(1)	$\rho(h) = \phi^h$; $\rho(1) = \theta/(1+\theta^2)$, $\rho(h) = 0$ for $h \geq 2$
Identification	ACF cuts off: MA; PACF cuts off: AR; both decay: ARMA
Criteria	AIC = $-2 \ln L + 2k$, BIC = $-2 \ln L + k \ln T$; the smallest wins
Forecast interval	$\hat{y}_{T+h} \pm 1.96 \sigma \sqrt{\sum_{j<h} \psi_j^2}$

What you need today (2/4): unit roots, ARIMA, SARIMA, evaluation

Quantity	Formula
ADF	$\Delta y_t = c + bt + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$; $H_0: \gamma = 0$ (unit root)
ADF, 5% critical values	about -2.87 (constant), -3.43 (constant and trend); reject if τ is smaller
KPSS	H_0 : stationarity; 5% critical values 0.463 (constant), 0.146 (trend); reject if larger
ARIMA(0,1,1)	$\Delta y_t = \varepsilon_t + \theta \varepsilon_{t-1}$; $\hat{y}_{T+h} = y_T + \theta \varepsilon_T$, $\alpha_{SES} = 1 + \theta$
Its error variance	$\sigma^2[1 + (h-1)(1+\theta)^2]$
AR(1) $\times$ SAR(1) _s	$z_t = \phi z_{t-1} + \Phi z_{t-s} - \phi \Phi z_{t-s-1} + \varepsilon_t$
Annual rate from z	$z_t = \Delta \Delta_s \ln y_t$; $\Delta_s \ln y_{T+1} = \Delta_s \ln y_T + \hat{z}_{T+1}$
Diebold–Mariano	$d_t = e_{1t}^2 - e_{2t}^2$, $DM = \bar{d}/(s_d/\sqrt{n})$; reject equal accuracy if $ DM > t_{0.975}(n-1)$

What you need today (3/4): volatility and several series

Quantity	Formula
GARCH(1,1)	$\sigma_{t+1}^2 = \omega + \alpha \varepsilon_t^2 + \beta \sigma_t^2$, $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$, $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$
Annual volatility	$\sqrt{q \bar{\sigma}^2}$, q = trading days a year
VaR 1%	$-(\mu + \sigma_{t+1} q_{0.01}(z))$; Student- t_ν : $q = t_\nu^{-1}(0.01) \sqrt{(\nu - 2)/\nu}$; Normal: -2.326
VAR(p)	$y_t = c + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t$, OLS equation by equation
Granger F	$[(RSS_R - RSS_U)/p] / [RSS_U / (T - Kp - 1)] \sim F(p, T - Kp - 1)$
Long-run effect	$\partial y / \partial x = \sum_j b_{yx,j} / (1 - \sum_j b_{yy,j})$
Engle-Granger, ECM	ADF on OLS residuals, critical value about -3.34 (two variables); $h_{1/2} = \ln 0.5 / \ln(1 + \gamma)$
Johansen	trace $-T \sum_{i>r} \ln(1 - \hat{\lambda}_i)$; test $r = 0, 1, \dots$ in turn; stop at the first non-rejection

What you need today (4/4): extensions

Quantity	Formula
ARFIMA(0, d , 0)	$\rho(1) = d/(1 - d)$, $H = d + 1/2$; stationary for $d < 1/2$, mean-reverting for $d < 1$
Local level	$F_t = P_t + \sigma_\varepsilon^2$, $K_t = P_t/F_t$, $a_{t t} = a_t + K_t(y_t - a_t)$, $P_{t+1} = P_t(1 - K_t) + \sigma_\eta^2$
Steady state	$\bar{P}/\sigma_\varepsilon^2 = (q + \sqrt{q^2 + 4q})/2$, $\bar{K} = \alpha_{SES}$, $q = \sigma_\eta^2/\sigma_\varepsilon^2$
Markov switching	$E(D_i) = 1/(1 - p_{ii})$, $\pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$
Machine learning	walk-forward validation only; compare with a simple benchmark
Values	$\chi_{0.95}^2$: 3.84 (1), 5.99 (2), 9.49 (4), 12.59 (6), 18.31 (10); $F_{0.95}(2, 250) = 3.03$

A1: an ARIMA(0,1,1) by hand [Solved]

Task

- ▣ $\Delta y_t = \varepsilon_t - 0.6 \varepsilon_{t-1}$, $\varepsilon_t \sim WN(0, 1)$; y_0 given, $\varepsilon_0 = 0$; today $y_T = 100$, $\hat{\varepsilon}_T = 1.5$.
- ▣ 1. Show that Δy_t is stationary and invertible, and compute $\rho_{\Delta y}(1)$.
- ▣ 2. Compute $\text{Var}(y_t)$ for $t = 10$ and $t = 100$ and say whether y_t is stationary.
- ▣ 3. Forecast y_{T+1} and y_{T+4} ; give the 95% interval for $h = 4$.
- ▣ 4. Give the weight of the equivalent exponential smoothing.
- ▣ Report: five numbers, an interval and one sentence.

Solution

- ▣ 1. An MA(1) is always stationary; root $1/0.6 = 1.667 > 1$: invertible; $\rho(1) = -0.6/1.36 = -0.441$
- ▣ 2. $y_t = y_0 + \varepsilon_t + 0.4 \sum_{j < t} \varepsilon_j$: $\text{Var} = 1 + (t-1)0.16$: 2.44 and 16.84: grows with t , not stationary
- ▣ 3. $\hat{y}_{T+h} = 100 - 0.6 \times 1.5 = 99.10$ for every h ; variances 1.00 and 1.48; interval [96.72, 101.48]
- ▣ 4. $\alpha = 1 + \theta = 0.4$
- ▣ Interpretation: the forecast is flat, like that of exponential smoothing, but its interval keeps widening: a unit root



A2: identifying the Romanian unemployment rate [Proposed]

Solution discussed in class

Task

Series	ADF τ	p	ADF crit. 5%	KPSS (crit. 0.463)
u_t	-1.52	0.524	-2.87	1.037
Δu_t	-8.12	< 0.001	-2.87	0.096

- Unemployment rate u_t (SA, %), January 2005 – August 2026; both tests with a constant. Δu_t ($T = 259$): ACF $-0.19, 0.11, 0.09, -0.07$; PACF $-0.19, 0.08, 0.13, -0.04$ (lags 1–4). Model: Seminar 3, A1; Seminar 2, A6.
- 1. State H_0 of each test and decide the order of integration of u_t .
- 2. Compute the band $\pm 1.96/\sqrt{T}$ and Ljung–Box $Q(4)$ from the four ACF values; decide at 5%.
- 3. Propose an ARIMA model for u_t .
- Report: two decisions, two numbers, a model.

A3: SARIMA output for Romanian GDP [Proposed]

Solution discussed in class

Task

Dependent: z_t	Coef.	SE	z	p
AR(1)	-0.083	0.106	-0.79	0.431
SAR(4)	-0.405	0.085	-4.79	< 0.001
SIGMASQ	10.41			
Ljung-Box $Q(8)$	18.14			0.006

- $z_t = \Delta\Delta_4 100 \ln \text{GDP}_t$, real GDP not seasonally adjusted, 2001Q2 – 2026Q2 ($T = 101$). Last values of z : 1.923 (2025Q2), -1.150 (2025Q3), -2.584 (2025Q4), -0.247 (2026Q1), -0.457 (2026Q2); annual growth $\Delta_4 100 \ln \text{GDP}$ in 2026Q2: -1.96%. Model: Seminar 4, A4.
- 1. Write the model equation and say which coefficients are significant.
- 2. Test whether the residuals are white noise ($\chi^2_{0.95}(6) = 12.59$).
- 3. Forecast z and the annual growth rate for 2026Q3.



A4: VAR output and Granger causality [Proposed]

Solution discussed in class

Task

	π_t equation	i_t equation
π_{t-1}	1.318 [21.58]	0.031 [0.48]
π_{t-2}	-0.342 [-5.52]	0.008 [0.13]
i_{t-1}	0.011 [0.19]	1.167 [19.60]
i_{t-2}	-0.012 [-0.22]	-0.219 [-3.81]
const.	0.112 [1.62]	0.063 [0.88]
R^2	0.97	0.97

- VAR(2) for Romanian annual inflation π and ROBOR 3M i (%), March 2005 – August 2026, $T = 258$; t in brackets. Without the lags of π , the RSS of the i equation rises from 89.81 to 92.20. Model: Seminar 6, A5.
- 1. Write the i equation.
- 2. Compute the Granger F statistic for “ π does not Granger-cause i ” and decide at 5% ($F_{0.95} = 3.03$).
- 3. Compute the long-run effect on ROBOR of a permanent 1 pp rise in inflation.

A5: GARCH output for the DAX and VaR 1% [Proposed]

Solution discussed in class

Task

DAX	μ	ω	α	β	ν
estimate	0.081	0.0206	0.097	0.895	7.08
SE	0.012	0.0049	0.011	0.012	0.617

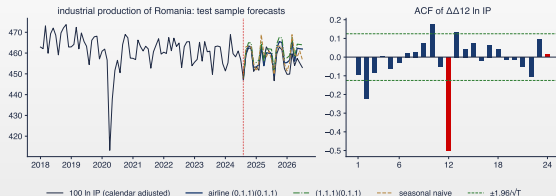
- GARCH(1,1)- t , daily DAX log returns in % since 2000 ($n = 6\,786$); last day (September 2026): $\sigma_T^2 = 0.662$, $\varepsilon_T = -1.698$; about 254 trading days a year; use $\nu = 7$: $t_7^{-1}(0.01) = -2.998$. Model: Seminar 5, A1 and A4.
- 1. Compute the persistence, the half-life and the long-run annual volatility.
- 2. Compute σ_{T+1}^2 .
- 3. Compute the one-day VaR 1% and compare it with the Normal VaR 1%.



B1: Box–Jenkins for Romanian industrial production [Solved]

- **Question:** which SARIMA model forecasts Romanian industrial production best, and does it beat the seasonal naive method?
- **Inputs:** industrial production (B–D), calendar adjusted, monthly, January 2005 – July 2026; test sample: the last 24 months
- **Tasks**
 1. Run ADF and KPSS on $100 \ln y_t$, its first difference and $z_t = \Delta \Delta_{12} 100 \ln y_t$.
 2. Plot the ACF of z_t and propose a model.
 3. Fit the airline model and SARIMA(1, 1, 1)(0, 1, 1)₁₂ on the training sample; compare AICc, BIC and Ljung–Box.
 4. Forecast the 24 test months with both models and with the seasonal naive method; compare RMSE and MAE.
 5. Interpretation: which model would you use to forecast, and why?
- **Report:** a table of tests, a table of two models, three RMSE and one sentence
- **Notebook:** section B1

B1: solution [Solved]



Model	AICc	BIC	LB(12) p	LB(24) p	test RMSE
airline $(0, 1, 1)(0, 1, 1)_{12}$	1304.2	1314.3	0.065	0.432	3.75
$(1, 1, 1)(0, 1, 1)_{12}$	1291.1	1304.5	0.789	0.972	5.29
seasonal naive					3.85

- Tests: ADF $\tau = -1.31$ ($p = 0.885$) and KPSS $0.523 > 0.146$ for the level; z_t : ADF -6.47 , KPSS 0.055 : $d = D = 1$; ACF of z_t : -0.09 , -0.22 at lags 1, 2, -0.50 at lag 12
- Airline: $\hat{\theta} = -0.205$, $\hat{\Theta} = -0.934$; the $(1, 1, 1)$ model: $\hat{\phi} = 0.574$, $\hat{\theta} = -0.828$, almost cancelling factors
- Interpretation: the $(1, 1, 1)$ model wins in sample (BIC, clean residuals) but loses out of sample; the airline model is slightly better than the seasonal naive method: use the airline model, and keep the seasonal naive method as the benchmark

B2: volatility of the BET since 2015 [Proposed]

- **Question:** is the Bucharest market calmer or more turbulent today than usual, and what does it mean for tomorrow's VaR 1%? Model: Seminar 5, B1.
- **Inputs:** BET daily log returns in %, January 2015 – September 2026 ($n = 2\,925$)
- **Tasks**
 1. Run the ARCH-LM test with five lags on the demeaned returns.
 2. Fit a GARCH(1,1) with Student- t innovations; report the persistence and the half-life.
 3. Compare the long-run, the sample and today's annualised volatility.
 4. Compute tomorrow's VaR 1% with the t and with the Normal quantile.
 5. Interpretation: is the BET calmer or more turbulent than usual today?
- **Report:** one test, five parameters, three volatilities, two VaR values and one sentence
- **Notebook:** section B2

B3: the Romanian and German 10-year yields [Proposed]

- **Question:** is the Romanian 10-year yield tied to the German one in the long run? Model: Seminar 7, B1 and A3.
- **Inputs:** monthly means of daily 10-year yields (%), January 2008 – September 2026 ($T = 224$)
- **Tasks**
 1. Run ADF on both yields and on their differences.
 2. Run the Engle–Granger test (Romania on Germany) and report $\hat{\beta}$.
 3. Run the Johansen trace and maximum-eigenvalue tests.
 4. Estimate the error-correction equation of each yield and compute the half-life.
 5. Interpretation: is the Romanian yield tied to the German one?
- **Report:** four ADF results, two cointegration tests, two adjustment coefficients and one sentence
- **Notebook:** section B3

C1: a forecast competition for Romanian inflation [Proposed]

- **Question:** which method would have forecast Romanian monthly inflation best over the last ten years, one and twelve months ahead?
- **Inputs:** the Romanian HICP (Eurostat), monthly, 2005–2026; Lecture 15 runs one SARIMA on one test sample
- **Tasks**
 1. List five competitors: seasonal naive, SES, SARIMA, ARFIMA, a ridge model with lags and month dummies.
 2. Describe a rolling-origin design: first origin, step, horizons, refits.
 3. Say how you would treat the tax changes of 2010, 2015 and 2025.
 4. Choose the error measures and the test for the comparison.
- **Report:** a one-page plan that a team could turn into its project
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to check the answers to Part A. The answer:
- ▣ (a) For the unemployment rate, ADF gives $p = 0.52$, so the unit root is rejected at 5%.
- ▣ (b) KPSS = 1.04 > 0.463 rejects, so the level of unemployment is stationary.
- ▣ (c) For the DAX, $\alpha + \beta = 0.992$, so a shock halves in $1/(1 - 0.992) = 128$ days.
- ▣ (d) The VaR 99% of the DAX for tomorrow is -2.31%.
- ▣ (e) Inflation Granger-causes ROBOR ($p = 0.036$), which proves that inflation causes the central bank to raise rates.
- ▣ (f) Ljung-Box $Q(12)$ on the residuals of an AR(1) x SAR(1) model has 12 degrees of freedom.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and the correct number (notebook, section C2).
 - ▶ 2. Report: six verdicts with one line of justification each.

Key takeaways

- Every exam problem has three parts: the formula with the numbers, the result with its unit and sign, the interpretation
- Tests: state H_0 first; ADF and KPSS have opposite nulls; Ljung–Box on residuals loses k degrees of freedom
- Forecasts by hand: write the model in levels of z_t , substitute the last values, then undo the differences
- A better in-sample model is not a better forecaster (B1); judge it on a test sample against a benchmark
- An AI answer is a draft: check the null hypotheses, the half-life, the VaR level and the degrees of freedom

After the seminar

- Lecture 15 gathers the course: the course map, one recap slide per chapter, the toolbox, Box–Jenkins on Romanian inflation, eight solved exam-type problems, the project
- Try the [Proposed] tasks on paper first, then check them in the notebook
- C1 can grow into a team project
- More practice: the [practice set](#) of 25 exam problems with numerical answers
- Reading: [Huang and Petukhina \(2022\)](#); [Hyndman and Athanasopoulos \(2021\)](#); [Box, Jenkins and Reinsel \(2008\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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